

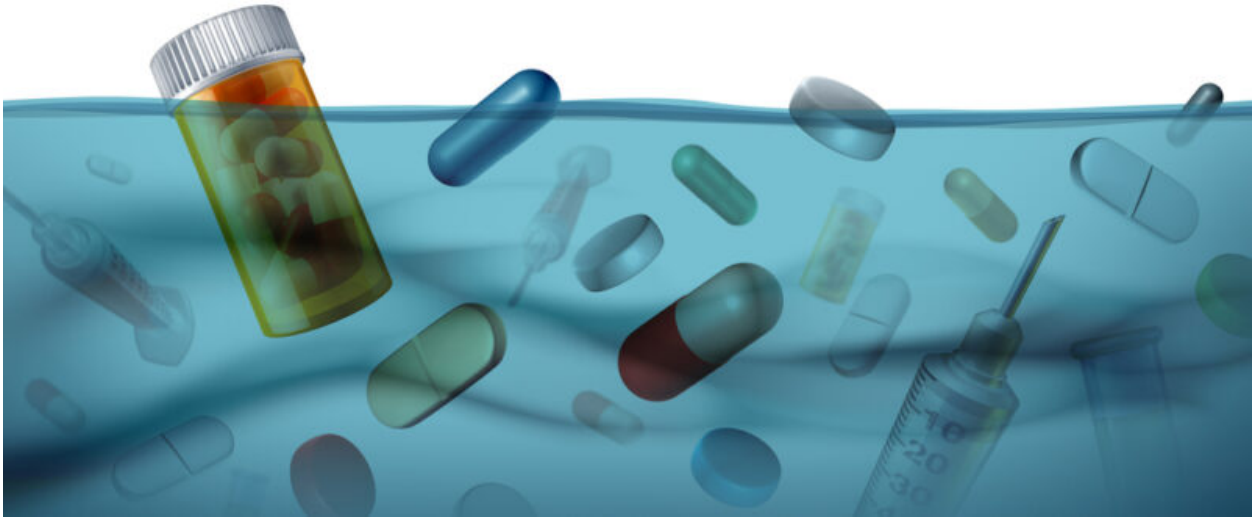
“Pharmaceutical Pollution in the Hudson River: An Educational Intervention"

By Hana Bauta

The Undergraduate Research Writing Conference 2023

Rutgers, The State University of New Jersey

Pharmaceutical Pollution in the Hudson River: An Educational Intervention



Submitted by:
Hana Bauta

Submitted to:
Ernest J. Tollerson III
Chairman of the Board of Directors
Hudson River Foundation

Date:
April 28, 2022
Scientific and Technical Writing
355:302:20
Professor Lena Sandberg-Golden

Abstract

Pharmaceutical pollution is a growing environmental concern that threatens the wellbeing of humans and wildlife across the world. The nationwide exponential growth of filled prescriptions is a contributory factor, worsening the natural consequences on local and global environments. On average, 55% of Americans take four prescription medications regularly and 75% Americans take at least one over-the-counter drug (Consumer Reports, 2017). Over two decades, the total number of prescriptions filled by all Americans, including adults and children, has climbed by 85%, even though the overall population of the United States has only expanded by 21% (Consumer Reports, 2017). With increasing drug use, the presence of pharmaceutical compounds in waterways is amplified by the improper widespread disposal of unused medications. Drugs such as antibiotics, hormonal agents, antidepressants, and painkillers have a direct route of exposure into major water sources from the flushing and tossing of medications. These eco-toxic chemical residues which bypass wastewater treatment filtration processes are then directly discharged into watersheds and estuaries teeming with life. As of 2014, the United States' bodies of water have tested positive for hundreds of various pharmaceuticals (Beek et al., 2016). The Hudson River has also demonstrated concerning levels of such contaminants. More precisely, the water pollutant levels in New York State's Lower Hudson Valley were recognized at alarmingly high and saturated levels. Municipal wastewater treatment plant outfalls near Rockland County and Westchester County in the Lower Hudson Valley tested positively for about 16 pharmaceutical drugs ranging from painkillers to antibiotics (Cantwell et al., 2018). Local and international pharmaceutical research has indicated disastrous implications to contaminated aquatic life and noted the creation of antibiotic-resistant superbugs. It is demanding to count on preventative source control methods rather than intervention to tackle this environmental problem.

This preventative solution will be accomplished through a pharmaceutical education initiative. The implementation of environmental health promotion practices through educational campaigning will allow for a community-based approach and widespread reach to household populations throughout Rockland and Westchester County. Infographic campaigns will relay information to the public communicating proper drug disposal through Drug Take-Back programs. Publicly distributed infographic brochures will be used as tools to illustrate and display valuable information regarding proper drug disposal and pharmaceutical facts. This plan will follow a model of success that was conducted as a Health Promotion campaign during the Covid-19 pandemic. To increase awareness and promote secondary vaccination doses, this model utilized visually informative infographic advertisements to educate a targeted audience. In a similar manner, this proposed solution will aim to foster the public's awareness of proper drug-take back disposal programs and educate on the hazards of drug contamination. Another model of success on the evaluation of social marketing for conservation will be used to highlight the successes of marketing campaigns on influencing behavior of a targeted community. Lastly, sample German and Finnish pharmaceutical campaigns will be used as models for graphic design, reflecting on organizational components for a comparable environmental solution. The goal of this plan is to raise awareness within the Rockland and Westchester communities and create a greater conversation surrounding this issue. Educating the public and influencing behavior related to pharmaceutical pollution will directly contribute to protecting the environment, wildlife, and local communities.

Table of Contents

	Page
Abstract……	**i**
Table of Contents……	**ii**
Table of Figures……...	**iii**
Introduction……	**1**
The Problem of Overprescription **……**	**1**
The Pollution of Waterways **……**	**2**
The Hudson River **……**	**3**
Environmental Consequences **……**	**5**
The Drug Take-Back Act **……**	**6**
Literature Review……	**8**
The Baltic Sea: Pharmaceutical Campaigns **……**	**8**
The Successes to Infographic Campaigning **……**	**9**
Conservation Social Marketing Campaign **……**	**11**
Plan……	**13**
An Educational Campaign **……**	**13**
Budget……...	**14**
Discussion……	**16**
A Call to Action **……**	**16**
References……	**17**

Table of Figures

	Page
Figure 1: United States Prescription Levels ……...	1
Figure 2: National Trends in Per Capita Pharmaceutical Spending …….	1
Figure 3: Number of Pharmaceuticals Detected Per Country …….....	2
Figure 4: Aquatic risks per Ecoregion for Ciprofloxacin …….....	3
Figure 5: Hudson River Valley Map …….....	3
Figure 6: Map of Hudson Sample Sites …….....	3
Figure 7: Frequency of Pharmaceutical Occurrence (%) …….....	4
Figure 8: Detected Pharmaceutical Concentrations …….....	4
Figure 9: CWPharma Campaign …….....	8
Figure 10: German Campaign Billboards …….....	8
Figure 11: SYKE Policy Brief…….....	9
Figure 12: Vaccine Clinic Appointment Data…….....	10
Figure 13: COVID-19 Vaccine Infographic…….....	10
Figure 14: Theory of Change Due to Factors …….....	12
Figure 15: Overview of Assorted Costs …….....	14
Figure 16: Percentage Breakdown of Assorted Costs…….....	15

Introduction

The Problem of Overprescription

The United States is a leader in global pharmaceutical consumption. On average, 55% of Americans take four prescription medications regularly and 75% of Americans take at least one over-the-counter drug (Consumer Reports, 2017) The overall amount of prescription drugs filled in the U.S. alone has increased from 2.4 billion in 1997 to 4.5 billion in 2016 (Consumer Reports, 2017). As indicated in Figure 1, prescription fills supplied to patients has increased by 85% between these years whereas the country's total population has increased by a mere 21%. The exorbitant usage of prescription drugs has surpassed the population's perimeters. Compared to peer nations, the U.S. spends more money on pharmaceutical spending than all other international countries. Figure 2 below illustrates the national trends in Per Capita Pharmaceutical Spending between the years 1980 to 2015. The United States skyrocketing growth in national spending on pharmaceutical drugs highlights a greater domestic concern. As the national overconsumption of pharmaceuticals increases, eco-toxic chemical compounds are undoubtedly being displaced back into the environment and surrounding ecosystems. Evidently, this exponential growth of drug usage within the United States has staggering consequences on the environment that are worsened with time. With more people continuing to take pharmaceutical drugs – antibiotics, hormonal agents, antidepressants, and painkillers – this problem is magnified.

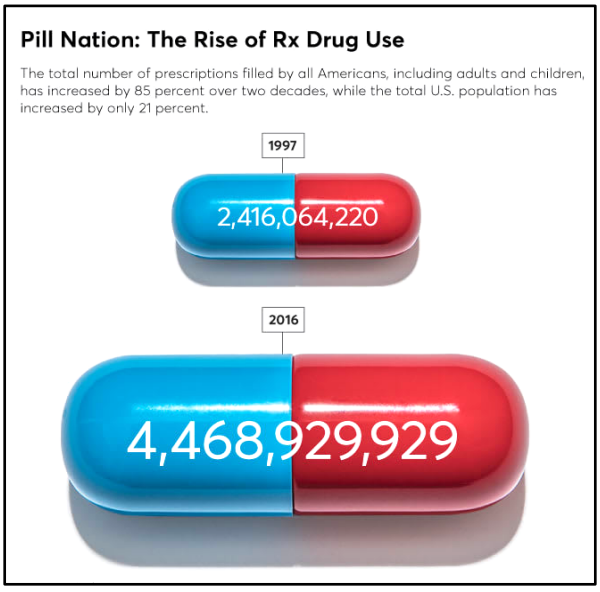


Figure 1: United States Prescription Levels (Consumer Reports, 2017)

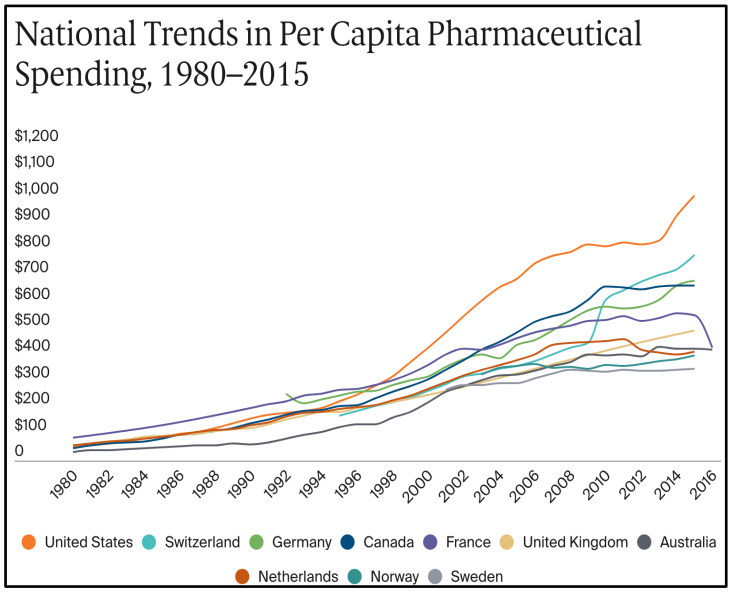


Figure 2: National Trends in Per Capita Pharmaceutical Spending (Organization for Economic Co-operation and Development, 2017)

The Pollution of Waterways

The pharmaceutical life cycle begins at the top of the production chain following factory-level administration and manufacturing. After local-level distribution to patients, the environmentally persistent eco-toxic compounds found in these prescription medications reenter the environment. These drugs have a direct pathway to main water supplies via human excretion and the improper flushing or tossing of medications (USGS, 2018). Unused or expired pharmaceuticals that are flushed and disposed of into sewage systems reemerge in the water supply network. Although wastewater goes through extensive filtering at treatment plants, a significant percentage of drug compounds can still make their way through into major water sources (Beek et al., 2016). Figure 3 demonstrated below reveals a global map highlighting the number of detected pharmaceutical compounds in various water sources throughout the world. In the United States, 101 to 200 different substances were detected in either surface, ground, tap or drinking water (Beek et al., 2016). It is evident that pharmaceutical water pollution is both a widespread environmental concern as well as a local one.

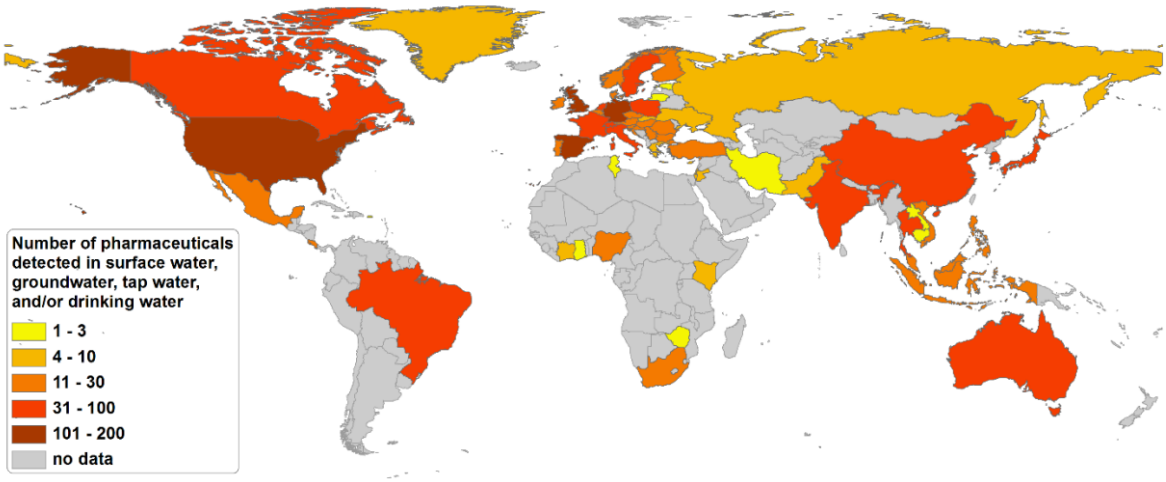


Figure 3: Number of pharmaceuticals detected in surface waters, groundwater, tap water, and/or drinking water per country (Beek et al., 2016)

According to research from the journal *Environmental Research Letters*, the harm related to ecological damage from the persistent compounds in ciprofloxacin — an antibiotic — was 10 to 20 times greater between the years 1995 to 2015 (Oldenkamp et al., 2019). Figure 4 pictured below, represents the increasing prevalence of this antibiotic and the worsening environmental, aquatic risks associated with the drug under a span of twenty-one years. The dangers of ciprofloxacin exposure from human consumption are predicted to be both progressively substantial and widespread (Oldenkamp et al., 2019). Ciprofloxacin is one of many other pharmaceutical drugs that will end up back in the main water supply. It is safe to assume that with time, pharmaceutical pollution will develop into a complex entity which permeates borders due to the interconnected nature of water. It is imperative that intervention and prevention practices be implemented on large-scale and small-scale levels as soon as possible to prevent the further exacerbation of pharmaceutical pollution.

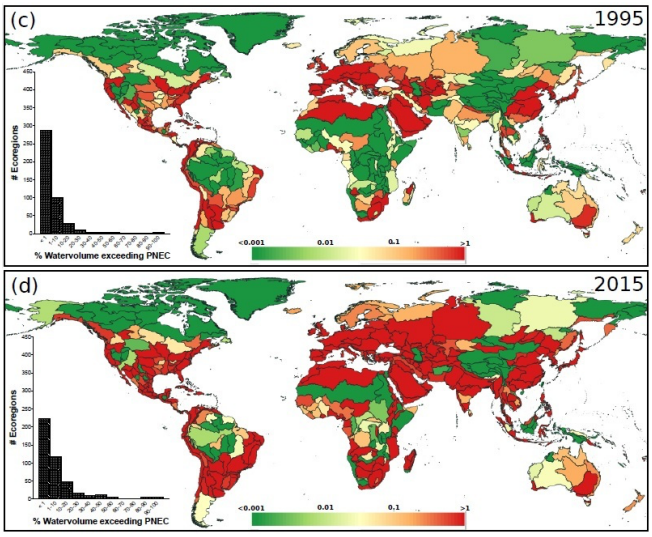


Figure 4: Aquatic risks per Ecoregion for Ciprofloxacin (Oldenkamp et al., 2019)

The Hudson River

Flowing throughout New York State is a vital water source and tidal estuary home to a variety of aquatic species: The Hudson River. The Hudson is situated adjacent to numerous, highly populated communities in New Jersey, New York City and New York State**.** Figure 5 shown below presents a map of the Upstate NY, Hudson River Valley region that is respectively divided by counties into lower, middle, and upper geographic regions. These populous counties all border the 300-mile length Hudson River from NYC and throughout Upstate New York. According to the U.S. Census, these ten adjacent Hudson counties have a large, combined population estimate of nearly 3,000,000 (2020). Due to the high population density bordering this river, the Hudson is exhibiting inevitable characteristics of human contamination. New York State is facing the repercussions of pharmaceutical water pollution.

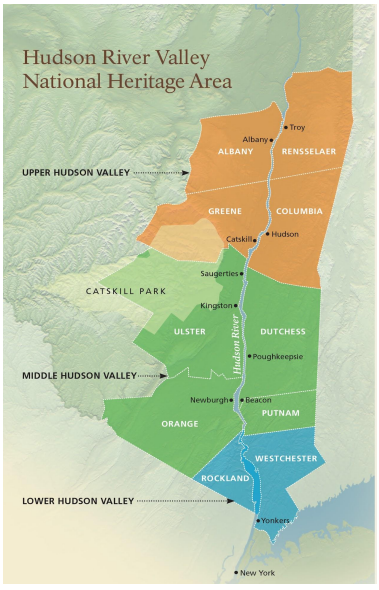


Figure 5: Hudson River Valley Map (National Park Service)

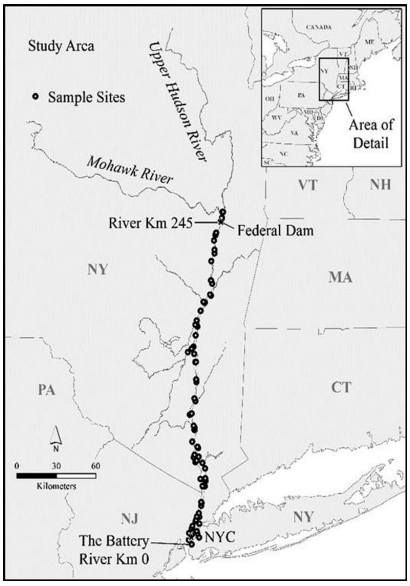


Figure 6: Map of Hudson Sample Sites (Cantwell et al., 2018)

As expected, alarming new research has found that the Hudson River is significantly polluted with a variety of prescribed pharmaceutical drugs. Water samples taken along the Hudson River indicated high compound levels near the sewage outfalls of several upstate New York municipalities (Krajick, 2018). This research – published in the Water Research Journal – claimed "16 highly prescribed pharmaceuticals were quantified along a 250 km transect of the Hudson River Estuary and New York Harbor" (Cantwell et al., 2018). Figure 6 demonstrated above displays the distinct locations of each water sample taken across the stretch of the Hudson River. Additionally, the 65 tested WWTP outfall sites were specified using distinct numeral descriptions which represent each sample's relation to its river kilometer position (RK). The water extractions were tested and analyzed for 16 common, human-use pharmaceutical compounds. Several of the positively observed drugs were categorized as antibiotics, analgesics, anticonvulsants, antihypertensives and diuretics.

Sewage outfalls near Upstate New York towns of Orangetown and Yonkers tested positive for nearly ninety percent of the examined compounds (Krajick, 2018). These two outfalls – in the close vicinity to Rockland and Westchester County respectively – are in the Lower Hudson Valley and situated across from one another, separated by the river. The Orangetown STP outfall was indicated as RK 41.8 whereas the Yonkers STP outfall was indicated by RK 28.2 for analysis. Figure 7 below, highlights the exorbitantly high frequency of pharmaceutical occurrence at the 28.2 RK and 41.8 RK marks compared to other Hudson-tested outfall RK sites. Subsequently, figure 8 demonstrates the varying concentrations of detected pharmaceuticals with respect to each tested location. Once again, RK 28.2 and RK 41.8 stand out as outliers with higher concentration levels than several other tested locations.

According to the United States Census Bureau, Westchester County and Rockland County have a combined population estimate of over 1.3 million people as of 2020. Additionally, Westchester County has a household population of 353,485 and Rockland County has a household population of 101,167 (US Census, 2020). This accounts for a combined total household population of 454,652. Since these two highly populated counties are situated alongside the Hudson River, it is a necessity that the local citizens are aware of this community-wide problem. Based on the supporting data, this proposal will take into account both Westchester and Rockland County as two sites where the plan of action for public intervention will be tested.

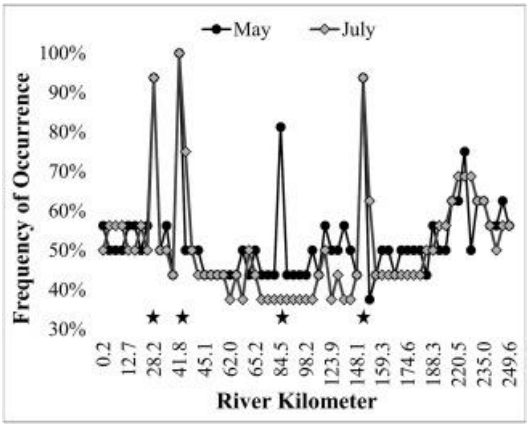


Figure 7: Frequency of Pharmaceutical Occurrence (%) (Cantwell et al., 2018)

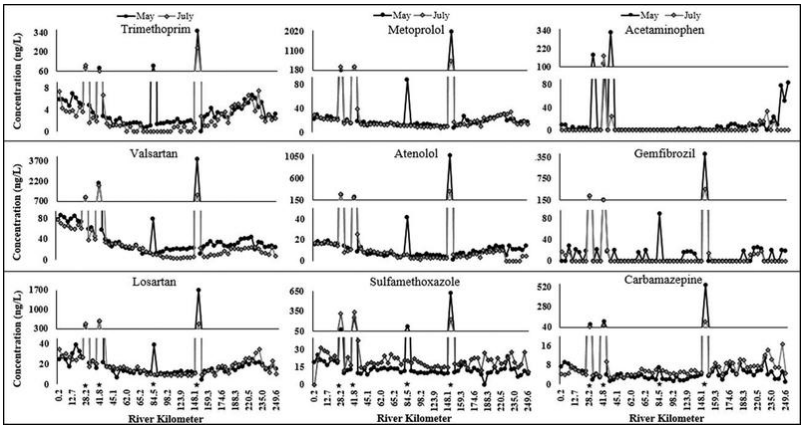


Figure 8: Detected pharmaceutical Concentrations (Cantwell et al., 2018)

Environmental Consequences

Water pollution in the Hudson River is detrimental to nearby watersheds and estuaries. In 2019, the Hudson was declared the second most endangered river in the United States (American Rivers, 2019). The Hudson River – being the East Coast's second largest estuary – provides essential habitat for endangered species such as Atlantic and Shortnose Sturgeon, as well as threatened species such as Banded Sunfish and Blanding's Turtles (Riverkeeper, 2019). The Hudson River is a vital source of life for various endangered species and a highly accessible, neighboring water source to the crowded Hudson communities. Thus, a varietal mix of pharmaceutical drugs reentering neighboring waterways is extremely concerning for both aquatic life and local communities.

The environmentally persistent compounds in pharmaceutical drugs are damaging to the natural composition of water and soil. Antibiotics – one of the most commonly prescribed medications – is used to treat bacterial infections by inhibiting the growth of bacteria. This drug has been found at alarming rates in water sources globally due to the excess drugs being disposed of into sewage systems. The environmental consequences of antibiotic runoff are catastrophic. They alter the characterization of microbes and bacteria, ultimately creating drug-resistant 'superbugs.' Antimicrobial resistance occurs when "bacteria, viruses, fungi and parasites change over time and no longer respond to medicines making infections harder to treat and increasing the risk of disease spread, severe illness and death" (WHO, 2021). According to the World Health Organization, Antimicrobial resistance is one of the "top 10 global public health threats facing humanity" (2021). Antimicrobial resistance is a dangerous phenomenon because drug-resistant microbes threaten the progress made in healthcare – causing effective medical and antibiotic treatments to no longer work. Exposure to antibiotic resistant microbes, from contact through swimming or water consumption, can prove both catastrophic and fatal to the general public.

There is a major public health concern regarding antibiotic-resistant bacteria within tri-state area water sources, with a distinct spotlight on the Hudson River. Local research on the Hudson River was conducted by CUNY Queens College and Columbia University. Culture-dependent samples discovered high levels of antibiotic-resistant bacteria in the water tested within close vicinity to sewage sites in Hudson River Estuaries (Young et al., 2013). Additional research conducted abroad discovered similar distinguishing characteristics. Laboratory experimentation in Tunisia examined antibiotic resistance among bacteria in wastewater from pharmaceutical industrial runoff (Tahrani et al., 2015). Isolated bacterial strains from this wastewater effluent demonstrated a 100% bacterial resistance to the antibiotic Amoxicillin. This evidence suggests that the creation of superbugs is both a global and local public health threat that needs to be publicly acknowledged.

Harmful pharmaceutical compounds extensively alter aquatic ecosystems and disrupt the biological makeup of native species which are exposed to the contaminated waters. Groundbreaking research demonstrates the catastrophic changes that occur within these ecological communities and vulnerable wildlife. A study conducted in Sweden analyzed the effects of pharmaceutical drugs on the blood plasma levels of fish. They found high blood concentrations of hormonal drug (levonorgestrel) and reduced fertility levels among effluent exposed fish (Fick et al., 2010). This alarming data depicts drug levels among fish equitable to

human concentration levels. Commonly prescribed hormonal drugs – such as progestin – which are made for human consumption can indirectly alter the biological levels of fish.

New research from the NY Cary Institute for Ecosystem Studies indicates that crayfish populations exposed to the antidepressant Citalopram – a psychoactive drug – develop more bold behavior by exposing themselves to predators for extended periods of time. SSRIs are effective in reducing anxiety and depression levels among humans; however, they also drastically increase levels of aggression and boldness in exposed aquatic species. This research indicates that "alterations to cray-fish behavior driven by SSRI exposure could lead to subsequent ecosystem-level effects" (Reisinger et al., 2021). With the increasing use of antidepressants there are also environmental consequences and irreversible changes. Antidepressants and antipsychotics play a role in altering food web dynamics by further disrupting natural ecosystem processes. There is a growing environmental problem regarding the extent to which pharmaceuticals alter the health and wellbeing of animals exposed to such eco-toxic compounds. There is an urgency to combat this pressing issue and put an end to the worsening progression of environmental degradation.

The Drug Take-Back Act

In 2019, the New York State Senate signed into law the "Drug Take Back Act." This new legislation holds big pharmaceutical manufacturers responsible for the cost, collection, relocation, and disposal of their used products. Both chain and mail-order pharmacies will be required to provide consumers with collection options in New York – such as drop boxes or prepaid mail-back envelopes – that would be at no cost to the consumer or pharmacy. Moreover, all rural, urban, and disadvantaged communities will have local access to the indicated collection services. These recently enforced measures will contribute to the reduction of medication misuse and ensure that the New York Hudson water reserves are safe from exposure to flushed drugs. Additionally, the implementation of this new law will contribute to the roll-out of many more drop-boxes throughout the New York area. It is important that the public is aware of these readily available disposal methods, otherwise, the value and effectiveness of such programs will be undermined if not utilized.

Studies have shown that Drug take-back programs can be effectively used as a preventative remedy to environmental contamination. The University of North Texas conducted research that analyzed the effectiveness of Drug disposal programs on reducing pharmaceutical concentrations in local water sources. To determine if Take-Back programs minimize drug residue levels in the environment, scientists monitored the concentration of hydrocodone in wastewater released from a treatment plant in Denton, Texas before and after a local take-back program. The collected data clearly suggests that these events were successful in contributing to a "reduction of pharmaceutical loading to the environment" (Stoddard & Huggett, 2015). The Denton Drug Disposal Day event had a significant impact on collecting and keeping harmful drugs out of the environment. This research provides noteworthy evidence on the success of Drug Take-Back Programs and showcases a direct correlation with reducing drug residue levels in wastewater sources.

Although there is growing research showcasing the success of drug disposal programs, the public is generally unaware of the harm of flushing medications and greatly uninformed of the

resources available to them. A survey of pharmacy patrons in New York suggests that “less than one third of patients receive information related to proper medication disposal” (Shealy et al., 2014). Patients at an outpatient pharmacy who completed a survey about medication disposal practices and beliefs indicated that “only 22.9% reported returning medication to a pharmacy for disposal, [and] less than 20% had ever been given advice about medication disposal by a healthcare provider” (Seehusen & Edwards, 2006). These separate studies both suggest that there is a role for patient education about the proper disposal of unused and expired medications. Lack of information and counseling on the proper disposal of medication increases the likelihood of these pharmaceutical compounds reentering the environment through damaging routes of exposure.

Literature Review

The Baltic Sea: Pharmaceutical Campaigns

The Baltic Sea – located in Northern Europe – is another major body of water that is exhibiting alarming quantities of micropollutants. The European Union (EU) has within recent years made strides in recognizing the impact of pharmaceutical pollution. In 2019, the EU Commission proposed a strategy to address and curtail pharmaceuticals in the environment (SYKE, 2019). There have been promotional campaigns conducted throughout the continent, with several EU member states coming together to dedicate resources to combat this problem and raise awareness on the issue. The 'Interreg Baltic Sea Region' is an EU-funded initiative that promotes collaboration in the Baltic Sea region. CWPharma – an Interreg Baltic Sea Region project – consists of "15 organizations, from seven countries in the Baltic region: Denmark, Estonia, Finland, Germany, Latvia, Poland, and Sweden…[whose] work is coordinated by the Finnish Environment Institute (SYKE)" (CWPharma, 2020).

Along with scientific research and intervention, the CWPharma Project has been responsible for crafting effective and engaging pharmaceutical campaigns. Figure 9 pictured below demonstrates the promotional advertisements developed by the project to educate the German public on proper drug disposal. This campaign was converted into a physical installation through advertisement billboards – as seen in Figure 10 – which were posted across Germany, targeting local consumers. The goal of this initiative was to raise awareness on the environmental impacts of pharmaceuticals and ultimately decrease the emissions of eco-toxic compounds in the Baltic Sea. The billboards emphasize the phrase "Gib der Natur nicht den Rest," "Entsorge Medikamente richtig" which translates to "Don't give nature the rest," "Dispose of medications properly." The simplistic text dialogue is accompanied with images of unfinished, leftover pharmaceutical medications. This international model highlights the intended message in a concise manner with supporting graphics, minimal text, and references to more information using a website address.



Figure 9: CWPharma Campaign (BMU)



Figure 10: German Campaign Billboards (SYKE)

Another initiative by the Finnish Environment Institute (SYKE) was the creation of a graphic-based policy brief from 2019. This document – Figure 11 – showcases the European Union (EU) strategic approach to diminishing the environmental consequences from pharmaceutical pollution. It is an educational tool intended to inform the reader on the different pharma routes to water exposure and encourage several different measures to combat the issue. Evidence-based research is paired with complementary, stylized data and visuals. There is a call to action directed towards the reader on the last page of the brochure which reads "Now is the time to act." Moreover, a website address is referenced as a closing remark to redirect the readers to a linked, educational European Union (EU) site on more information related to the topic. This campaign serves as a framework for graphic design and a reference to the organization of information. Comparatively, my plan will be more graphic based with less textual support for easy reading comprehension across Westchester and Rockland County. Avoiding overpowering text and relying on graphical support is important to conveying a message without overwhelming the reader and appealing to all reading levels.

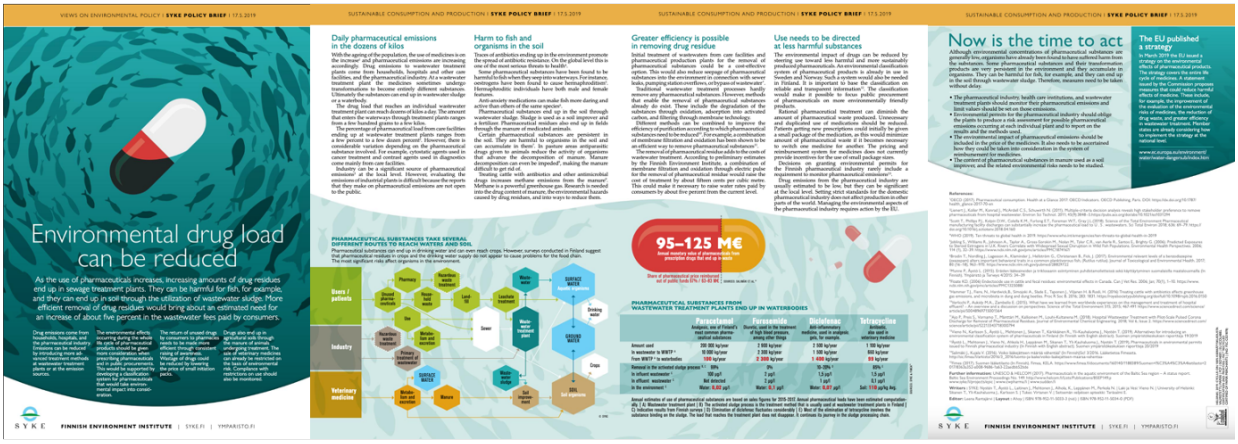


Figure 11: SYKE Policy Brief (Nystén et al., 2019)

The Successes to Infographic Campaigning

One of the most prevalent demonstrations of a successful education program is the Covid-19 effort that effectively promoted public health measures to the general population. During the height of the pandemic, the CDC and WHO launched a global health campaign to educate the public and boost vaccination rates. After the release of the initial Moderna and Pfizer vaccine doses, the number of individuals returning for second shots decreased due to vaccine hesitancy and misinformation. A team of researchers analyzed the success of public health ads in the form of infographics, on encouraging follow-up vaccination. The study – organized after the release of the second Covid-19 dose – focused on creating infographics for public consumption to explain the purpose and importance of a second dose. The results represented in Figure 12 indicate that "the proportion of COVID-19 vaccine recipients returning for a second inoculation grew after COVID-19 vaccine infographics were distributed to first-time vaccine recipients" (Crutcher & Seidler, 2021). Following the distribution periods of the flyers, more people were returning for a 2nd dose within this specific time frame. Importantly, this demonstrates the successes of the overall project in influencing human behavior toward an intended end goal.

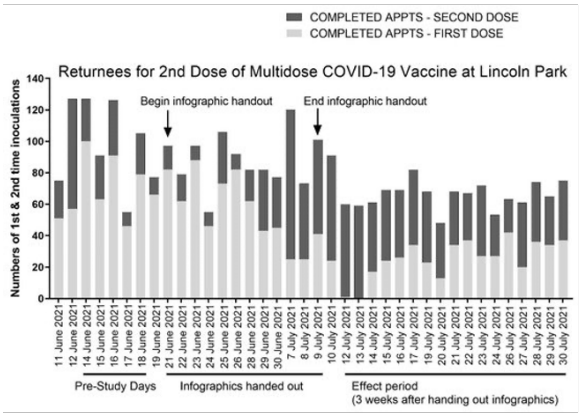


Figure 12: Vaccine Clinic Appointment Data (Crutcher& Seidler, 2021)

The study design of this research focused on a local sample of vaccinated individuals from the Lincoln Park vaccination clinic in Los Angeles, CA. The educational Covid-19 infographic was made available to first-time vaccine recipients. Seen in Figure 13 below, it is displayed with easy-to-understand wording and pictograms conveying the purpose and function of the second vaccine. There was an increase in returns for second doses following the release of the graphically displayed information. Tools, such as infographics, are modes of information that are easily distributed and convey straightforward knowledge to uninformed individuals. This real-world example demonstrates the visible success in health promotion through extended communication and educational campaigns. Moreover, this study proves that administering infographic flyers is an effective method of educating the public and getting the message out to the community. Concise text dialogue and related graphics in the flyers worked to successfully persuade patrons to return for a second shot. This project highlights the successes of infographic campaigning as a method of raising awareness and influencing behavior among a specific community. A solution to reducing pharmaceutical pollution will be reflective of the successes of public education through print media and the distribution to an appropriate audience.

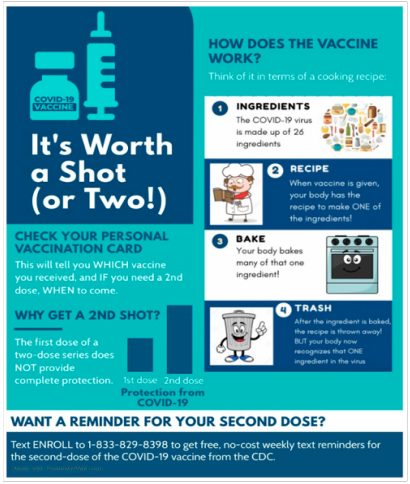


Figure 13: COVID-19 Vaccine Infographic (Crutcher& Seidler, 2021)

Conservation Social Marketing Campaign

Social marketing campaigns are an approach to engage people and educate them on critical issues with the goal of altering their behavior. This method has been used in the public health field to address issues as diverse as health, mental health, prevention, harm reduction, and conservation. This methodology borrows heavily from traditional marketing and advertising, concepts that have previously existed. Companies have leveraged these approaches to sell goods and services to consumers, and they have been extremely effective in altering consumer behavior. This is evidenced by increased advertising and marketing throughout every sphere of commercial space, such as billboards, TV commercials, internet ads, social media advertising, and many other forms of media where products and services are advertised to potential customers.

Social marketing is an effective method in altering negative, maladaptive behaviors and socializing behaviors that have positive outcomes. Conservationists – from a study in *Conservation Biology* – used this approach to design a cost-effective campaign to aid the recovery of endangered wildlife species. Social marketing was used to educate the public, encourage the preservation of sensitive environments with the goal of reducing human footprint in areas where endangered species resided. Researchers utilized this approach for the benefit of biodiversity as well as protecting endangered habitats. In their study, the team evaluated the effectiveness of this methodology by assessing the influence of human behavior and conservation outcomes (Salazar et al., 2018). A qualitative evaluation method was used to assess the long-term impact of the '*1998 Rare Pride'* campaign on the island of Bonaire. The campaign was designed to increase the population an endangered parrot species (i.e., Aasona Barbadensis or Lora) in the Dutch Caribbean. Researchers utilized the data from the campaign to develop an overall theory of change to explain changes in the percentage of the parrot/Lora population. The study utilized participant interviews with veterinarians, biologists, educators, individuals working in tourism, conservation (i.e., rangers, forestry staff), public sector such as government officials, media representatives, and local population, to collect data on status of the Lora population.

The study surveyed the groups to collect information on three domain areas such as: current knowledge of the Lora population (i.e., population estimates) perceived risks, and explanations regarding changes to the habitat or human intervention that led to the overall reduction of the Lora population and their respective habitats. To test the theory of change based on social marketing as a vehicle for behavioral change, the researchers developed potential causes for population change of the Lora and asked the participants to outline factors that lead to relative risk. This information was used to determine the perceived relative risk of habitat loss and degradation based on human interventional factors. The researchers identified 29 factors that influenced the Lora population over 20 years, including economic changes and conservation efforts in the last few years. The analysis of the results and data identified five factors and possible causal pathways that influenced the parrot/Lora population since 1997 (Salazar et al., 2018). Those include a change in the Lora's habitat and breeding grounds (reduction of overall habitat in the island), environmental laws, ecotourism, environmental education in schools, and social media campaigns. As seen in Figure 14 below, further analysis and identification of key

factors that had the greatest behavioral impact include: (1) Environmental laws, (2) Social Marketing Campaign, and (3) Environmental Education in Schools.

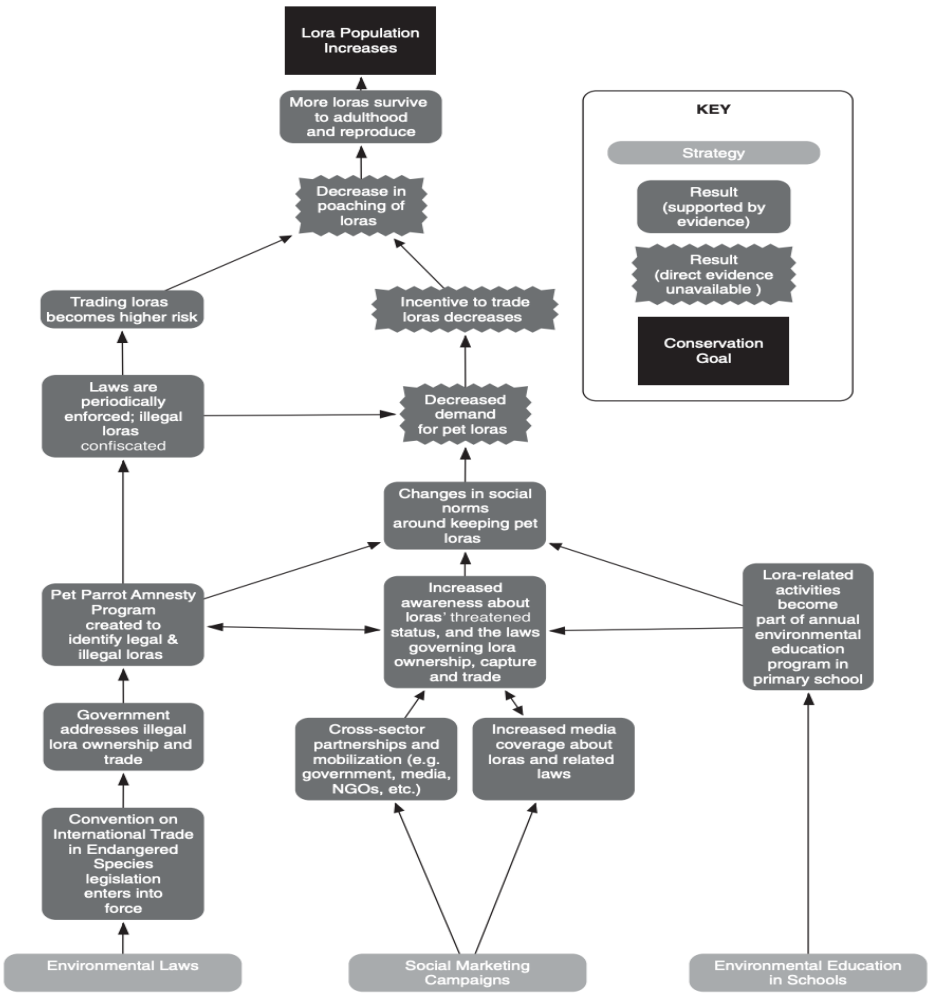


Figure 14: Theory of Change Due to Three Factors (Salazar et al., 2019)

Social marketing, alongside education, and laws were the three critical factors that ultimately changed the fate of the parrot/Lora population in the island of Bonaire. Laws served as formal agents of enforcement that served as a deterrent in engaging in Lora habitat destruction for commercial purposes and sale of the parrots for the pet trade. Additionally, education of young children in schools served as a social reinforcer to preserve local habitats. Lastly, the Social Media campaign was the key factor to alter adult behavioral change in the protection and preservation of the Lora habitat and environment. This evaluation demonstrates the effectiveness of this methodology to alter behavior and act as a public health tool. All in all, this research supports the framework of this proposal as a proven approach to enhance behavior change.

The Plan

An Educational Campaign

Ensuring that patients are properly educated and instructed on disposal methods is vital to the prevention of irreversible environmental harm. Medications that are disposed of into the trash, toilet, or sink, risk the safety of others and damage the environment. Environmental Health Promotion will play a key component in combating this growing problem. The plan of action to promote source control of pharmaceuticals is through the implementation of county-based education programs. It is necessary to educate New York residents in the lower Hudson Valley about the possible environmental effects of what they stock in their medicine cabinets, as well as encourage the act of proper drug disposal. This includes the promotion of pharmaceutical drug take-back programs and awareness of the hazards of improper drug disposal from flushing or pouring unwanted, unused, or expired medications down the drain. Infographics are powerful digital tools that illustrate and display valuable information through easy-to-understand visuals and graphics. They will be vital resources in relaying proper drug disposal and pharmaceutical facts to the public in Westchester and Rockland County. This includes raising public awareness of the presence of safe, take-back programs that are offered as pharmacy-based collections, law enforcement-based collections, community event style programs, and drug mail-back programs. All in all, it is important that the public is aware of the various disposal options available to them.

The targeted audience that will receive the public education pamphlets are household units that reside in Rockland and Westchester Counties. Each of the combined 454,652 household units (Census, 2020) will receive infographic brochures with permission from the two county municipalities. Each household in these two Hudson communities will be informed via targeted door-to-door mailing that includes information about the Drug Take-Back programs. To encourage adoption, the plan will be initially discussed with local government leaders in both municipalities to highlight the issue. The goal of presenting this topic in a public town hall forum includes gaining the necessary support for the project and having the approvals to disseminate the media campaign. The infographic flyers will be visually-focused with a clear message advising the readers to avoid drains – toilets and sinks – for pharmaceutical disposal. The handout pamphlets and posters will include the phrase "#MedicationEducation" to encourage people to search for more information online and further expand their knowledge on this subject. Additionally, social media will be used to further amplify the message about the public health education program by encouraging public officials and others to use "#MedicationEducation" in their postings. The public education campaign approach includes the development of a website with the same phrase addressed as "www.MedicationEducation.com." The website will act as a repository of information on this issue. Redirecting the audience to more information through hashtags and websites will further raise awareness and create a conversation about pharmaceutical pollution.

The Budget

The costs associated with this program include funding necessary to support the application of Pharmaceutical Drug education programs throughout the New York counties of Westchester and Rockland. The Hudson River Foundation's "New York City Environmental Fund" will specially assist in covering expenses associated with the program's countywide roll-out. The estimated total budget of the plan is rounded out to be approximately $92,000 which includes the assortment of costs related to creating and distributing the educational campaigns. Figure 15 below is a table overview of the different costs associated with each component of the program.

Graphic Designer	$100 or more per design
Computer Programmer	Between $1000 and $5000 per website $50 per month for maintenance
Printing	$0.04 per pamphlet Around $50 per 1000 printed brochure leaflets
Website Domain	$4,295
Distribution Labor	Leaflet distributors paid per 1000 flyers for $130 or $15 per hour Volunteer environmentalists

Figure 15: Overview of Assorted Costs

These educational campaigns will require designers who can convert the essential information into a visually appealing and digestible formats for consumers. Compensation will be required for the designers' work. This includes the cost of hiring a graphic designer for each brochure design as well as a computer programmer to create an aesthetically pleasing educational site for the accompanying online website. Purchasing the domain *www.MedicationEducation.com* should also be considered. Individuals who search up the phrase "#MedicationEducation" will be relocated to this informative site with links to more resources and information. In addition, the cost of black-and-white and colored printing is a significant price point which is calculated per 1000 printed brochures with reference to the household quantities in Westchester and Rockland County. Colored brochures are important to the ensuring the success of the educational campaign by graphically separating important informational components and catching the readers' attention. The cost of labor and transportation which includes the price of hiring distributors should also be taken into consideration. The cost of distribution of pamphlets and posters throughout the two counties will be another principal price point. However, these costs can be reduced with the help of volunteer organizations from environmentalist clubs and communities.

Figure 16 highlights the general breakdown of the overall costs. As depicted, a majority of the budgetary funds will be allocated towards printing and distribution costs.

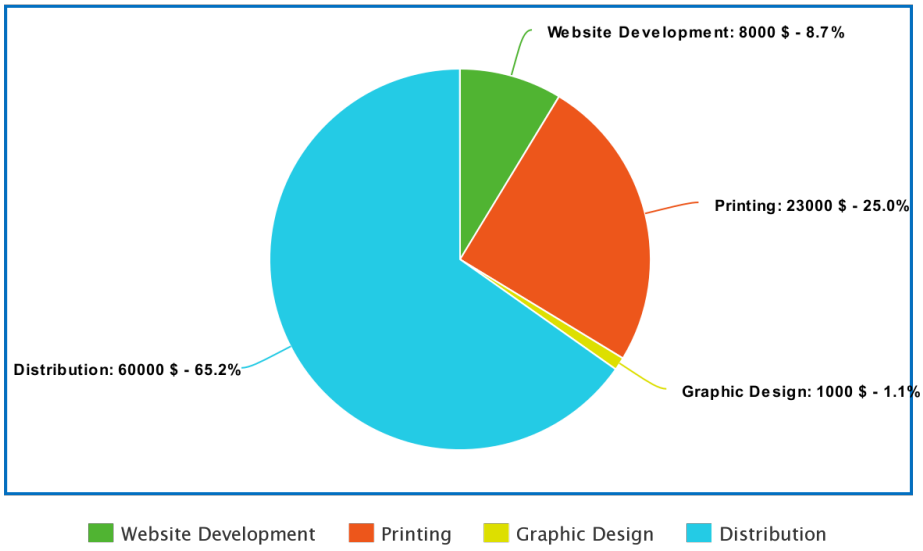


Figure 16: Percentage Breakdown of Assorted Costs

Discussion

A Call to Action

Freshwater is a limited resource that is vital to life on Earth. Since the Hudson River is a significant source of freshwater for the Hudson River Estuary, it greatly demands stewardship. As the second largest estuary on the East Coast, this river is home to countless endangered and native species. There is a growing concern surrounding water quality as a result of increasing pharmaceutical pollution across the globe – concretely threatening the well-being of the Hudson River. Since the impact of eco-toxic pharmaceuticals is a new and growing concern due to rapidly increasing drug use, promoting awareness of the issue can contribute to adequate prevention. Without prevention and source control, pharmaceuticals will continue to adversely affect New Jersey and New York's environment. The communities inhabiting the vicinity of the Hudson and the aquatic life within these waters can face damaging consequences to their wellbeing.

Hopefully, the assorted scientific research and detailed plan outlined above have convinced you of the urgency of this situation. I hope that you agree that there needs to be a prompt solution to this worsening problem and that you can see the promise in this proposal. Together we share a common goal of improving and protecting the local environment and community. This plan of action is a low-cost initiative that will influence great change and undoubtedly ensure the Hudson River's safekeeping through the prevention of pharmaceutical pollution and the spread of public awareness. It is imperative that this issue be addressed now before the environmental degradation caused by eco-toxic pharmaceutical compounds reaches a pinnacle and causes irreversible damage to the Hudson River and Hudson Valley communities.

I hope that you strongly consider financially supporting my plan of action to implement county-wide educational campaigns in Westchester and Rockland County. I am confident promoting valuable information to the public on the hazards of improper drug disposal and alternative Drug Take-Back resources will successfully raise awareness on a community-level and directly influence people to change their habits for the better. Additionally, starting on a small scale will allow for larger implementation overtime with the optimistic end goal for promoting large-scale awareness across the United States. All in all, this educational campaign will be one step closer to protecting the environment from harmful pharmaceutical substances by keeping the waters clean. I am excited to further explore this project and hope to make a change in the Hudson Valley community by spreading awareness on this pressing environmental issue.

References

American Rivers. (2019). *American Rivers Announces America's most Endangered Rivers of 2019*. American Rivers. Retrieved April 28, 2022, from https://www.americanrivers.org/conservation-resource/american-rivers-announces-americas-most-endangered-rivers-of-2019/

Aus der Beek, T., Weber, F. A., Bergmann, A., Hickmann, S., Ebert, I., Hein, A., & Küster, A. (2016). Pharmaceuticals in the environment: Global occurrences and perspectives. *Environmental toxicology and chemistry*, *35*(4), 823–835. https://doi.org/10.1002/etc.3339

BMU [Online image]. (2020). “Don't give nature the rest” A campaign from 2020. https://www.bmuv.de/richtig-entsorgen-wirkt/kampagne-gib-der-natur-nicht-den-rest

Cantwell, M. G., Katz, D. R., Sullivan, J. C., Shapley, D., Lipscomb, J., Epstein, J., Juhl, A. R., Knudson, C., & O'Mullan, G. D. (2018). Spatial patterns of pharmaceuticals and wastewater tracers in the Hudson River Estuary. *Water Research, 137*, 335-343. https://doi.org/10.1016/j.watres.2017.12.044

Consumer Reports. (2017). Do Americans take too many prescription medications? *Media Room*. Retrieved March 1, 2022, from https://www.consumerreports.org/prescription-drugs/too-many-meds-americas-love-affair-with-prescription-medication

Crutcher, M., & Seidler, P. M. (2021). Maximizing Completion of the Two-Dose COVID-19 Vaccine Series with Aid from Infographics. *Vaccines*, *9*(11), 1229. https://doi.org/10.3390/vaccines9111229

CWPharma. (2020, September 3). *The Baltic Sea region’s waste management for unused*

medicines must be improved [Press release]. https://www.cwpharma.fi/en-US/News_and_events/The_Baltic_Sea_regions_waste_management_(58477)

Fick, J., Lindberg, R. H., Parkkonen, J., Arvidsson, B., Tysklind, M., & Larsson, D. G. (2010). Therapeutic levels of levonorgestrel detected in blood plasma of fish: results from screening rainbow trout exposed to treated sewage effluents. *Environmental science & technology*, *44*(7), 2661–2666. https://doi.org/10.1021/es903440m

Finnish Environment Institute. (2021, June 21). *Actions to reduce pharmaceutical emissions in the Baltic Sea region*. SYKE. Retrieved April 28, 2022, from https://www.syke.fi/en-US/Current/Actions_to_reduce_pharmaceutical_emissio(61036)

Krajick, K. (2018). Researchers map high levels of drugs in the Hudson River. *Columbia Climate School.* Retrieved March 9, 2022, from https://news.climate.columbia.edu/2018/02/19/high-levels-of-drugs-found-dissolved-in-many-parts-of-hudson-river/

Lubick N. (2010). Drugs in the environment: do pharmaceutical take-back programs make a difference?. *Environmental health perspectives*, 118(5), A210–A214. https://doi.org/10.1289/ehp.118-a210

National Park Service. [Online image]. Hudson River Valley National Heritage Area. https://www.hudsonrivervalley.com/regions

Nystén, T., Äystö, L., Laitinen, J., Mehtonen, J., Alhola, K., Leppänen, M., & Perkola, N. (2019, January 17). *Environmental drug load can be reduced*. SYKE. Retrieved April 28, 2022, from http://hdl.handle.net/10138/301744

Oldenkamp, R., Beusen, A. H., & Huijbregts, M. A. (2018). Aquatic risks from human

pharmaceuticals—modelling temporal trends of Carbamazepine and ciprofloxacin at the Global Scale. *Environmental Research Letters*, *14*(3), 034003. https://doi.org/10.1088/1748-9326/ab0071

Organization for Economic Co-operation and Development. (2017). National Trends in Per Capita Pharmaceutical Spending, 1980–2015. [Online image]. The Commonwealth Fund. https://www.commonwealthfund.org/chart/2017/national-trends-capita-pharmaceutical-spending-1980-2015

Reisinger, A. J., Reisinger, L. S., Richmond, E. K., & Rosi, E. J. (2021). Exposure to a common antidepressant alters crayfish behavior and has potential subsequent ecosystem impacts. *Ecosphere,* 12(6). https://doi.org/10.1002/ecs2.3527

Riverkeeper. (2019, April 16). *Hudson River named one of America's most endangered rivers of 2019*. Riverkeeper. Retrieved April 28, 2022, from https://www.riverkeeper.org/news-events/news/preserve-river-ecology/hudson-river-named-one-of-americas-most-endangered-rivers-of-2019/

Salazar, G., Mills, M. and Veríssimo, D. (2019). Qualitative impact evaluation of a social marketing campaign for conservation. Conservation Biology, 33: 634-644. https://doi.org/10.1111/cobi.13218

Seehusen, D. A., & Edwards, J. (2006). Patient practices and beliefs concerning disposal of medications. *Journal of the American Board of Family Medicine*, *19*(6), 542–547. https://doi.org/10.3122/jabfm.19.6.542

Shealy, K. M., O'Day, P., & Eagerton, D. H. (2014). The Needs and Opportunities for Medication Disposal Programs. *Journal of Pharmacy Technology*, 30(5), 147–150. https://doi.org/10.1177/8755122514545519

Stewart, H., Malinowski, A., Ochs, L., Jaramillo, J., McCall, K., 3rd, & Sullivan, M. (2015). Inside Maine's Medicine Cabinet: Findings From the Drug Enforcement Administration's Medication Take-Back Events. *American journal of public health*, *105*(1), e65–e71. https://doi.org/10.2105/AJPH.2014.302207

Stoddard, K.I., & Huggett, D.B. (2015) Wastewater Effluent Hydrocodone Concentrations as an Indicator of Drug Disposal Program Success. *Bulletin of Environmental Contamination and Toxicology,* 95, 139–144. https://doi.org/10.1007/s00128-015-1574-9

Tahrani, L., Soufi, L., Mehri, I., Najjari, A., Hassan, A., Van Loco, J., Reyns, T., Cherif, A., & Ben Mansour, H. (2015). Isolation and characterization of antibiotic-resistant bacteria from pharmaceutical industrial wastewaters. *Microbial pathogenesis*, *89*, 54–61. https://doi.org/10.1016/j.micpath.2015.09.001

U.S. Census Bureau. QuickFacts. (2020). Retrieved April 1, 2022, from https://www.census.gov/quickfacts/fact/table/westchestercountynewyork/HSD410219

USGS. (2018, June 6). *Pharmaceuticals in Water*. U.S. Geological Survey. Retrieved April 6, 2022, from https://www.usgs.gov/special-topics/water-science-school/science/pharmaceuticals-water

World Health Organization. (n.d.). *Antimicrobial resistance*. Retrieved March 9, 2022, from https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance

Young, S., Juhl, A., & O'Mullan, G.,D. (2013). Antibiotic-resistant bacteria in the Hudson River Estuary linked to wet weather sewage contamination. *Journal of Water and Health, 11*(2), 297-310. http://dx.doi.org/10.2166/wh.2013.131